

Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist

July 2022 - Present

Nevada National Security Site

- Developed a YOLO-based object-detection model in Pytorch that achieved a 80% MAP evaluation on an in-house thermal-image vehicle dataset. The model was trained on AWS with four GPUs by leveraging Pytorch's DataParallel.
- Designed an anomaly-detection model in Pytorch using an LSTM variational-autoencoder to detect component failure in a pulse-power x-ray machine. The model accurately predicted the location of a component failure with 87% accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with Pytorch to develop a deblurring model for a pulse-powered x-ray machine. The model was used applied by the machine operators and reduced machine configuration time by 30%.
- Took initiative to develop the company's first deep-learning interface that wrapped Pytorch.
 - Introduced a standardized approach for training, validating and testing models that ensured reproducibility and model validity.
 - Ensured that code written by all colleagues could seamlessly be integrated together.
 - Supported multi-gpu training with the use of Pytorch's DataParallel.
 - Implemented dry-run checks to ensure the safe execution of a program.
- Maintainer of my group's AWS EC2 instance. Ensured the health of the instance, user-accounts and GPU functionality.
- Sole developer of a Django web interface, which wrapped Neo4j. The interface gave non-technical staff the ability to query and update the company's Neo4j graph-database without needing to know any query syntax.

Technical Skills

Programming Languages: Python, SQL, Bash, Cypher, Lua, $\LaTeX$

Libraries: Pytorch, Scikit-Learn, Pandas, Scipy, SQLAlchemy, Matplotlib, Plotly, Streamlit, Django

Developer Tools: Git, AWS, Unix/Linux, Vim, Tmux, Neo4j

Education

Auburn University - Ph.D.

June 2022

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S.

December 2015 / January 2018

Major: Applied Mathematics

Publications

- 1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). <https://doi.org/10.1007/s40072-022-00258-6>
- 2: Chen, L., Eisenberg, N. Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term. J Theor Probab (2024). <https://doi.org/10.1007/s10959-023-01302-4>